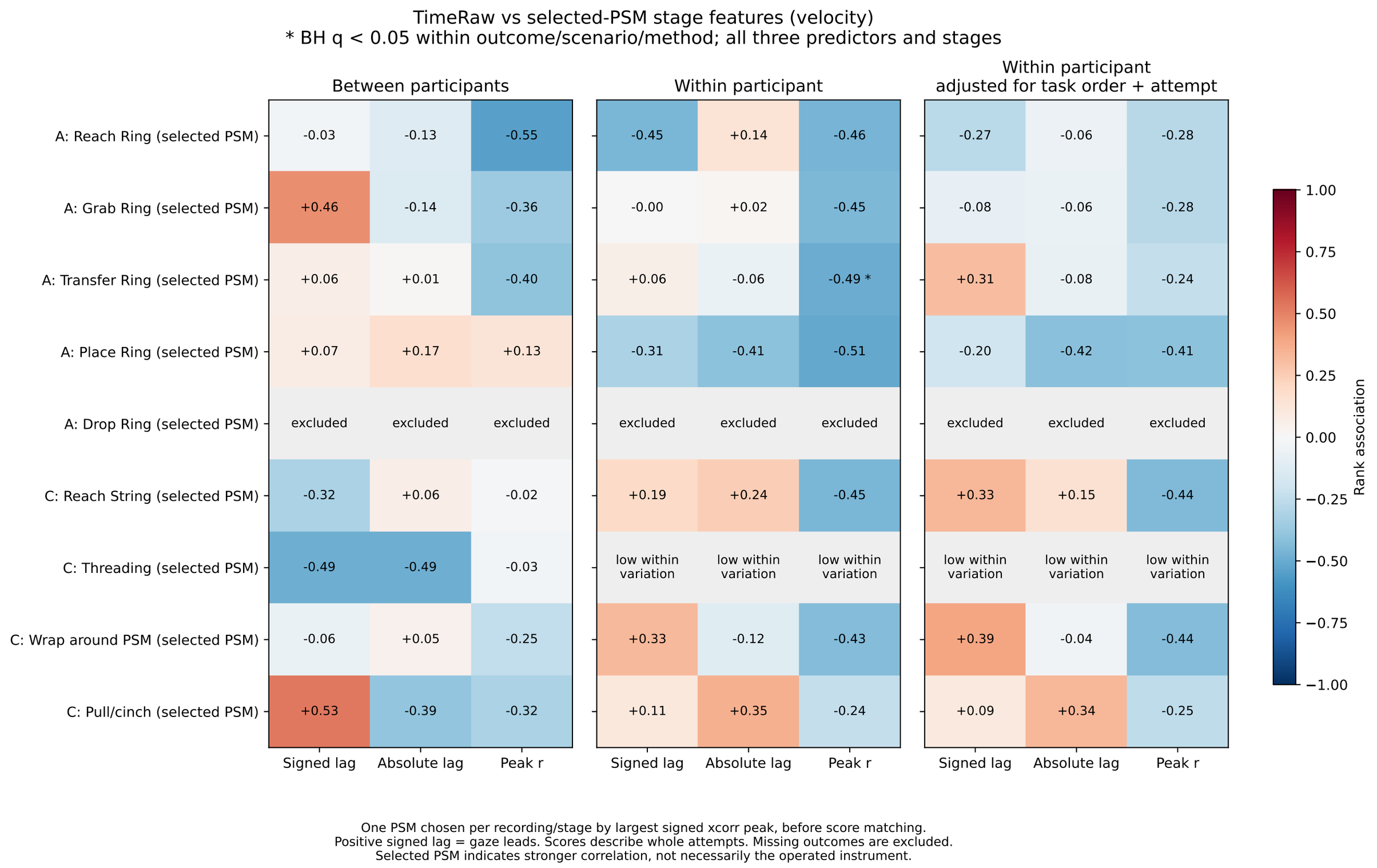
TimeRaw vs selected-PSM stage features (velocity)
* BH q < 0.05 within outcome/scenario/method; all three predictors and stages
Between participants
Within participant
Within participant
adjusted for task order + attempt
A: Reach Ring (selected PSM)
A: Grab Ring (selected PSM)
A: Transfer Ring (selected PSM)
A: Place Ring (selected PSM)
A: Drop Ring (selected PSM)
C: Reach String (selected PSM)
C: Threading (selected PSM)
C: Wrap around PSM (selected PSM)
C: Pull/cinch (selected PSM)
-0.03 -0.13 -0.55
-0.45 +0.14 -0.46
-0.27 -0.06 -0.28
+0.46 -0.14 -0.36
-0.00 +0.02 -0.45
-0.08 -0.06 -0.28
+0.06 +0.01 -0.40
+0.06 -0.06 -0.49 *
+0.31 -0.08 -0.24
+0.07 +0.17 +0.13
-0.31 -0.41 -0.51
-0.20 -0.42 -0.41
excluded excluded excluded
excluded excluded excluded
excluded excluded excluded
-0.32 +0.06 -0.02
+0.19 +0.24 -0.45
+0.33 +0.15 -0.44
-0.49 -0.49 -0.03
low within variation
low within variation
low within variation
low within variation
low within variation
low within variation
-0.06 +0.05 -0.25
+0.33 -0.12 -0.43
+0.39 -0.04 -0.44
+0.53 -0.39 -0.32
+0.11 +0.35 -0.24
+0.09 +0.34 -0.25
Signed lag
Absolute lag
Peak r
Signed lag
Absolute lag
Peak r
Signed lag
Absolute lag
Peak r
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Rank association
One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.